25901341 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCIAA201 - SERVICES & SPECIFICATIONS I

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **10** questions

Each question carries **2** marks

1. What is sound transmission loss? [K] / [CO2]
2. What is wood wool? [K] / [CO2]
3. Differentiate between natural and artificial light. [U] / [CO1, CO4]
4. What is illuminance? [U] / [CO1, CO4]
5. Define reflection of sound. [K] / [CO2]
6. What is task-based lighting planning? [U] / [CO1, CO4]
7. Define circuit protection. [K] / [CO1, CO5]
8. What is glass wool? [K] / [CO2]
9. Suggest fire protection measures for a room with high fire load. [U] / [CO3]
10. What happens when an electrical circuit is overloaded? [K] / [CO1, CO5]

11.	Apply fire-retardant treatment to wooden furniture in interiors.	[A] / [CO3]
12.	Why is daylight important for sustainable design?	[A] / [CO1, CO4]

[2x10 = 20]

Part B

Short Essay

Answer any **6** questions

Each question carries **5** marks

13.	Describe the diffusion of light and materials used for diffused lighting.	[U] / [CO1, CO4]
14.	Write about casing, capping and batten wiring methods.	[U] / [CO1, CO5]
15.	What are the factors that influence sound absorption?	[K] / [CO2]
16.	Explain the safety regulations to be followed in residential electrical installations.	[U] / [CO1, CO5]
17.	Explain the physical and psychological effects of good lighting in interiors.	[A] / [CO1, CO4]
18.	Write an account about the 'building wiring system'.	[U] / [CO1, CO5]
19.	Apply the principles of fire protection while planning the interior layout of a small commercial office.	[A] / [CO3]
20.	Explain the sound insulation of the floor and ceiling.	[U] / [CO2]

[5x6 = 30]

Part C

Essay

Answer any **2** questions

Each question carries **10** marks

21.	Explain fire detection and alarm systems used in residential and commercial buildings. Discuss the working of smoke detectors and sprinkler systems.	[U] / [CO3]
22.	Explain major factors to be considered while designing lighting for various interior spaces such as classrooms, offices, and residences.	[A] / [CO1, CO4]
23.	Explain various types of artificial lighting and their applications in interior spaces with suitable examples.	[A] / [CO1, CO4]

[10x2 = 20]